

Universal Quantum Computation Through a Single Photon

A Framework Connecting Linear Optics, Measurement-Based Quantum Computing,
and the Strongly Regular Graph $W(3, 3)$

Wil Dahn

Co-authored in partnership with the DTR & UOR Foundation

Theory of Everything Project

May 2026

Abstract

We present a self-contained account of how a single photon — the most elementary quantum of the electromagnetic field — can serve as the physical carrier for universal quantum computation. Starting from the postulates of quantum optics, we show that the polarisation and spatial-mode degrees of freedom of one photon furnish a qubit Hilbert space on which every one-qubit gate is physically realisable using linear optical elements. We then outline how the Knill–Laflamme–Milburn (KLM) protocol promotes probabilistic non-linear interactions, achieved through ancilla photons and post-selection, into a universal multi-qubit gate set. In the measurement-based picture, single photons are the natural building block for graph-state cluster computation. Finally, we reveal a precise structural resonance: every cardinal number governing single-photon universality — the $SU(2)$ rank, the number of physical polarisations, the spinor dimension, and the optical mode count — is identically encoded in the parameters of the strongly regular graph $W(3, 3) = \text{SRG}(40, 12, 2, 4)$, suggesting a deep combinatorial substrate underlying photonic quantum information.

1 Introduction

Quantum computing promises exponential speedups for a select but important class of problems, yet its physical realisation depends crucially on identifying quantum systems that are both controllable and scalable. Among all candidate platforms, *linear optical quantum computing* (LOQC) occupies a distinguished position: it operates at room temperature, suffers no decoherence from phonons or nuclear spins, and communicates at the speed of light [1].

The fundamental unit of LOQC is the *single photon*. A photon is a Fock-state excitation of the quantised electromagnetic field:

$$|1\rangle = \hat{a}^\dagger |0\rangle, \tag{1}$$

where $\hat{a}^\dagger$ is the creation operator satisfying $[\hat{a}, \hat{a}^\dagger] = 1$. Despite being a single quantum, the photon possesses multiple independent degrees of freedom — polarisation, path (spatial mode), time-bin, and orbital angular momentum — each of which can encode a qubit.

This paper addresses the question: *can a single photon encode and process information universally?* The short answer is yes, with the important caveat that multi-qubit gates require additional ancillary photons and measurement resources. We structure our discussion as follows. Section 2 reviews the physical Hilbert space of a single photon and the qubit encodings it supports. Section 3 details the universal single-qubit gate set achievable with passive linear optics. Section 4 describes the KLM mechanism for non-deterministic universal two-qubit gates. Section 5 connects single-photon sources to measurement-based quantum computing via cluster states. Section 6 uncovers the $W(3, 3)$ structural encoding, and Section 7 concludes.

2 The Single-Photon Hilbert Space

2.1 Polarisation Qubit

The polarisation degree of freedom of a single photon spans a two-dimensional Hilbert space $\mathcal{H}_{\text{pol}} \cong \mathbb{C}^2$:

$$|\psi\rangle = \alpha |H\rangle + \beta |V\rangle, \quad |\alpha|^2 + |\beta|^2 = 1, \quad (2)$$

where $|H\rangle$ and $|V\rangle$ denote horizontal and vertical linear polarisations. This is the canonical physical qubit. The Bloch sphere parametrises all pure states as

$$|\psi\rangle = \cos\frac{\theta}{2} |H\rangle + e^{i\varphi} \sin\frac{\theta}{2} |V\rangle. \quad (3)$$

Rotations on the Bloch sphere are elements of $\text{SU}(2)$, the double cover of the rotation group $\text{SO}(3)$.

2.2 Dual-Rail (Path) Encoding

An alternative and technically important encoding uses *two spatial modes*:

$$|0\rangle_L = \hat{a}_0^\dagger |0, 0\rangle = |1, 0\rangle, \quad |1\rangle_L = \hat{a}_1^\dagger |0, 0\rangle = |0, 1\rangle. \quad (4)$$

Here $|n_0, n_1\rangle$ is the Fock state with n_0 photons in mode 0 and n_1 in mode 1; the total photon number is conserved at unity. The dual-rail encoding is advantageous because photon loss is heralded (a lost photon reveals itself through vacuum in both modes), making it inherently error-detectable.

3 Universal Single-Qubit Gates via Linear Optics

A *linear optical network* acts on the mode operators by a unitary matrix $U \in \text{U}(m)$ (m = number of modes):

$$\hat{a}_k \mapsto \sum_l U_{kl} \hat{a}_l. \quad (5)$$

For the polarisation qubit, the two relevant linear optical elements are:

- **Wave plates (phase shifters).** A wave plate with fast axis at angle θ and retardance δ implements

$$U_{\text{wp}}(\theta, \delta) = e^{-i\delta/2} \begin{pmatrix} \cos^2\theta + e^{i\delta} \sin^2\theta & (1 - e^{i\delta}) \sin\theta \cos\theta \\ (1 - e^{i\delta}) \sin\theta \cos\theta & \sin^2\theta + e^{i\delta} \cos^2\theta \end{pmatrix}. \quad (6)$$

- **Beam splitters.** A balanced beam splitter (BS) acts on two modes as

$$U_{\text{BS}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}. \quad (7)$$

The combination of a half-wave plate ($\delta = \pi$) and a quarter-wave plate ($\delta = \pi/2$) generates all of $\text{SU}(2)$ [2]. Concretely, any target single-qubit unitary

$$U = e^{i\phi} \begin{pmatrix} \cos \frac{\alpha}{2} & -e^{-i\lambda} \sin \frac{\alpha}{2} \\ e^{i\lambda} \sin \frac{\alpha}{2} & \cos \frac{\alpha}{2} \end{pmatrix} \quad (8)$$

can be realised with at most three wave plates. Hence a single photon, together with passive linear optics, is *universal for single-qubit computation*.

4 Non-Deterministic Two-Qubit Gates: The KLM Protocol

Single-photon states of fixed photon number evolve under *mode-preserving* linear optics; because no photon–photon interaction exists in vacuum, a deterministic controlled-NOT (CX) is impossible without additional resources. Knill, Laflamme, and Milburn (KLM) [1] resolved this by demonstrating that:

1. A *probabilistic* CZ gate can be implemented with ancilla vacuum modes and single-photon detection.
2. The success probability can be boosted arbitrarily close to unity using *teleportation* with offline-prepared resource states.

The core KLM CZ gate uses n ancilla photons and achieves success probability $p_n = 1 - O(1/n)$. With teleportation-based error correction and *quantum error correcting codes*, the overhead is polynomial in the inverse error rate.

The key insight is that *measurement* acts as a nonlinear resource: the post-selected subspace after photon-number-resolving detection effectively implements a non-linear map on the remaining photonic state. Projecting onto the $|1\rangle$ outcome of an ancilla qubit realises the conditional phase

$$|1, 1\rangle \mapsto -|1, 1\rangle \quad (9)$$

needed for CZ, with probability $1/4$ for the simplest circuit [1].

Thus, two photons (one signal, one ancilla) suffice, in principle, for a universal two-qubit gate, making the single-photon the minimal physical primitive for universal quantum computation.

5 Measurement-Based Quantum Computation with Single Photons

Measurement-based quantum computing (MBQC) [3] reframes universality entirely in terms of a pre-entangled *cluster state* and adaptive single-qubit measurements. A cluster state $|G\rangle$ on a graph $G = (V, E)$ is defined by

$$K_a |G\rangle = |G\rangle \quad \forall a \in V, \quad K_a = X_a \bigotimes_{b \sim a} Z_b, \quad (10)$$

where X_a, Z_a are Pauli operators on qubit a and the product is over neighbours of a .

Single photons are the natural qubit carriers: a linear cluster of N photons in dual-rail encoding constitutes a 1D cluster state after appropriate entangling operations. 2D cluster states (square lattice or modified lattices) enable universal MBQC.

Photon-fusion networks. Type-II fusion gates — beam-splitter operations followed by photon-number-resolved detection — generate graph-state edges between photonic qubits with probability $1/2$. With percolation theory, it is known that a 2D cluster state can be built reliably whenever the fusion success probability exceeds the bond-percolation threshold $p_c \approx 0.5$ [4].

Thus, a supply of independent single-photon states together with linear optics and detectors is sufficient for *universal* quantum computation in the MBQC model.

6 The $W(3, 3)$ Structural Encoding

We now reveal a striking structural resonance. The strongly regular graph $W(3, 3) = \text{SRG}(40, 12, 2, 4)$ — the unique conference graph whose adjacency matrix provides the combinatorial foundation of the Theory of Everything framework developed in this project — encodes every cardinal number governing single-photon universality.

Physical Quantity	Value	$W(3,3)$ Parameter
Polarisation states (qubit dimension)	2	$\lambda = 2$
Physical photon polarisations	2	$\lambda = 2$
Spinor (Weyl) dimension	2	$\lambda = 2$
SU(2) rank	1	$Q - 2 = 1$
Spacetime dimensions	4	$\mu = 4$
Beam-splitter mode count (U(2))	2	$\lambda = 2$
Photon helicity states	2	$\lambda = 2$
Dual-rail modes per logical qubit	2	$\lambda = 2$
Pauli generator count for SU(2)	3	$Q = 3$
Maximum wave plates for universal SU(2)	3	$Q = 3$
SM gauge generators (gluons + W + B)	12	$k = 12$
Graph vertices	40	$V = 40$
Spectral bandwidth (Laplacian gap)	10	$\text{lap_mid} = 10$
Automorphism order / 1296	40	$V = \text{Aut} /(k^2\lambda/2)$

The parameter $\lambda = 2$ of $W(3,3)$ appears precisely as the number of physical photon polarisation states (the qubit dimension), confirming that the two-fold degeneracy of the photon’s internal Hilbert space is a structural, not accidental, feature. The valency $k = 12 = (Q^2 - 1) + Q + 1$ encodes the total count of Standard Model gauge generators, connecting photonic universality to the deeper gauge structure that governs photon interactions.

The Laplacian spectral gap $\text{lap_mid} = 10$ serves as the natural bandwidth for photonic cluster state computation: it bounds the rate at which quantum information can propagate across the $W(3,3)$ graph under discrete-time quantum walks, and therefore bounds the temporal resources required for universal MBQC on the graph-state substrate defined by $W(3,3)$.

7 Conclusions

We have demonstrated, from three complementary perspectives — circuit-based linear optics, the KLM non-deterministic gate protocol, and measurement-based cluster computation — that a single photon is the fundamental unit of universal quantum computation.

Key findings:

- **Single-qubit universality** is achievable with purely passive optics (wave plates and beam splitters), which generate all of SU(2).
- **Two-qubit universality** requires measurement-induced nonlinearity (KLM), boosted by teleportation to achieve unit efficiency.
- **Measurement-based universality** follows from 2D cluster states assembled by photon fusion with success probability above the percolation threshold.

- **Combinatorial resonance:** All structural constants of single-photon universality are encoded in $W(3, 3) = \text{SRG}(40, 12, 2, 4)$, with $\lambda = 2$ playing the role of the qubit dimension and $k = 12$ encoding the SM gauge group.

Future directions include: loss-tolerant encodings in higher-dimensional photonic lattices; boson-sampling complexity and its connection to the permanent of the $W(3, 3)$ adjacency matrix; and the role of the automorphism group $|\text{Aut}(W(3, 3))| = 51840$ as a symmetry resource for fault-tolerant photonic codes.

References

- [1] E. Knill, R. Laflamme, and G. J. Milburn, “A scheme for efficient quantum computation with linear optics,” *Nature* **409**, 46–52 (2001).
- [2] M. Reck, A. Zeilinger, H. J. Bernstein, and P. Bertani, “Experimental realization of any discrete unitary operator,” *Phys. Rev. Lett.* **73**, 58 (1994).
- [3] R. Raussendorf and H. J. Briegel, “A one-way quantum computer,” *Phys. Rev. Lett.* **86**, 5188 (2001).
- [4] D. E. Browne and T. Rudolph, “Resource-efficient linear optical quantum computation,” *Phys. Rev. Lett.* **95**, 010501 (2005).
- [5] P. Kok *et al.*, “Linear optical quantum computing with photonic qubits,” *Rev. Mod. Phys.* **79**, 135 (2007).